Network Traffic Analysis Investigation Report

1. Executive Summary

This report documents a comprehensive network traffic analysis conducted using Wireshark. The primary objective of the investigation was to analyze captured network traffic, identify communication patterns, detect suspicious network behavior, and validate potential threats using threat intelligence resources.

The investigation focused on DNS traffic analysis, TCP communication review, endpoint identification, packet stream inspection, and suspicious activity detection. Through systematic analysis of the captured traffic, valuable insights were obtained regarding network communications and potential security risks.

This exercise demonstrates practical skills commonly required in Security Operations Centers (SOC), Threat Hunting teams, and Blue Team environments.

2. Investigation Objectives

The objectives of this investigation were:

- Analyze captured network traffic using Wireshark
- Examine DNS communication activity
- Investigate TCP-based communications
- Identify active network endpoints
- Review packet streams for suspicious behavior
- Validate indicators using threat intelligence
- Document findings and security observations
- Develop practical packet analysis skills

3. Scope of Analysis

The investigation focused on the following network activities:

- DNS Queries and Responses
- TCP Communications
- Network Endpoints
- Top Communicating Hosts
- Packet Stream Analysis
- Threat Intelligence Validation

The analysis was performed on a packet capture (PCAP) file using Wireshark's traffic inspection capabilities.

4. Tools Used

Tool	Purpose
Wireshark	Packet capture analysis and protocol inspection
VirusTotal	Domain reputation and threat validation
TCP Stream Analysis	Session reconstruction and communication review
Endpoint Statistics	Identification of communicating hosts

5. Investigation Methodology

The investigation followed a structured approach similar to standard SOC investigation procedures.

Phase 1: Initial Traffic Review

The packet capture file was opened in Wireshark and examined to identify the overall network activity present within the capture.

Initial review activities included:

- Protocol identification
- Traffic volume assessment
- Communication pattern observation
- Host activity review

The purpose of this phase was to establish an understanding of normal and potentially abnormal network behavior.

Phase 2: DNS Traffic Analysis

DNS traffic was analyzed to determine which domains were being queried by network hosts.

The following activities were performed:

- DNS query inspection
- Domain resolution review

- Identification of unusual domain requests
- Threat intelligence validation of suspicious domains

DNS traffic analysis is important because attackers often rely on domain-based communication to deliver malware, establish command-and-control channels, or redirect users to malicious websites.

Several domain lookups were observed during the investigation. Domains that appeared suspicious were further investigated using VirusTotal to determine their reputation and risk level.

Phase 3: TCP Communication Analysis

TCP traffic was examined to understand communication between internal and external systems.

The analysis included:

- TCP session identification
- Connection establishment verification
- Handshake validation
- Communication flow analysis

The TCP three-way handshake process was observed and verified:

1. SYN packet transmitted
2. SYN-ACK response received
3. ACK packet returned

This confirmed successful connection establishment between communicating hosts.

The analysis also provided visibility into how data was exchanged between systems and allowed further investigation of suspicious sessions.

Phase 4: Endpoint Analysis

Endpoint analysis was conducted using Wireshark's statistical tools to identify active hosts participating in network communications.

The objectives included:

- Identifying internal devices
- Identifying external systems
- Reviewing communication frequency
- Detecting unusual external connections

Endpoint analysis provided valuable insight into communication patterns and highlighted systems generating significant network activity.

This phase is critical in threat hunting because unusual communication between endpoints may indicate malicious activity or unauthorized access.

Phase 5: Top Endpoints Investigation

The Top Endpoints feature was used to identify hosts responsible for generating the highest volume of traffic.

Activities performed included:

- Ranking hosts by traffic volume
- Reviewing communication frequency
- Investigating high-volume systems
- Identifying abnormal network behavior

High-volume traffic does not necessarily indicate malicious activity; however, it often serves as an important starting point for identifying anomalies.

The analysis provided additional visibility into network utilization and communication patterns.

Phase 6: TCP Stream Analysis

The Follow TCP Stream feature was used to reconstruct complete communication sessions.

This analysis allowed:

- Inspection of communication contents
- Session reconstruction
- Packet sequence review
- Detailed traffic investigation

Following TCP streams provides a clearer understanding of how hosts communicate and can reveal suspicious commands, data transfers, or unusual communication patterns.

The reconstructed streams were reviewed for indicators of suspicious or unexpected activity.

6. Suspicious Activity Investigation

Throughout the investigation, several areas of interest were examined.

DNS Investigation

- Multiple domain lookups were observed
- Suspicious domains were identified
- Reputation checks were conducted through VirusTotal

TCP Investigation

- External communications were reviewed
- TCP sessions were analyzed
- Packet streams were reconstructed

Endpoint Investigation

- Active hosts were identified
- Communication patterns were reviewed
- External IP addresses were examined

Traffic Volume Investigation

- High-volume endpoints were reviewed
- Communication behavior was analyzed
- Potential anomalies were assessed

The investigation did not rely solely on packet inspection. Threat intelligence validation was incorporated to provide additional context regarding observed network indicators.

7. Key Findings

The following findings were identified during the analysis:

Finding 1: DNS Activity Observed

Numerous DNS queries were identified throughout the packet capture. Several domains required additional validation to determine their reputation and legitimacy.

Finding 2: External Communications Present

Multiple communications between internal and external hosts were observed, indicating active network interaction with outside systems.

Finding 3: TCP Sessions Successfully Established

TCP analysis confirmed successful communication between multiple hosts through the standard TCP three-way handshake process.

Finding 4: Endpoint Visibility Achieved

Endpoint analysis successfully identified active hosts and provided insight into communication frequency and behavior.

Finding 5: Stream Analysis Improved Investigation Depth

TCP stream reconstruction enabled deeper visibility into network communications and improved understanding of session behavior.

Finding 6: Threat Intelligence Enhanced Validation

VirusTotal reputation checks provided additional context for evaluating suspicious domains and indicators observed during the investigation.

8. Security Recommendations

Based on the investigation findings, the following recommendations are provided:

Monitoring Recommendations

- Continuously monitor DNS activity
- Investigate unusual domain requests
- Monitor outbound network communications

Threat Detection Recommendations

- Integrate threat intelligence feeds
- Monitor suspicious external IP communications
- Establish network traffic baselines

Security Operations Recommendations

- Utilize packet analysis during incident investigations
- Conduct regular traffic reviews
- Implement SIEM monitoring for enhanced visibility

User and Infrastructure Recommendations

- Maintain updated security controls
- Review firewall policies regularly
- Implement network segmentation where appropriate
- Monitor high-volume network activity

9. Conclusion

This investigation demonstrated the practical application of network traffic analysis using Wireshark. Through DNS analysis, TCP communication review, endpoint investigation, and stream reconstruction, valuable insight was obtained regarding network behavior and communication patterns.

The combination of packet analysis and threat intelligence validation enabled a more effective assessment of potentially suspicious activity. The investigation strengthened essential cybersecurity skills related to traffic analysis, threat hunting, network monitoring, and incident investigation.

The project successfully achieved its objectives and provided hands-on experience with methodologies commonly used by SOC Analysts, Blue Team operators, and Incident Responders.

10. Analyst Note

This report was prepared as part of a practical cybersecurity project to demonstrate network traffic analysis, packet inspection, threat investigation, and security monitoring skills using Wireshark and threat intelligence resources.

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